Sparta data audit (D2): data flow

ANCP Group M. Planned flow for the H&M public dataset; the transactions file is not loaded yet, so no counts or results are shown.

1 Sources

H&M Personalized Fashion Recommendations (Kaggle)

articles.csv: product type, colour, section, index group
customers.csv: hashed ID, age, club status
transactions_train.csv: date, customer, article, price, channel

Access: Kaggle account, rules accepted
Licence: competition rules, non-commercial research and education
Status: files pending

Google Trends (optional)

Menswear search interest, India, CSV export; joined on week in step 4 if used
Public, aggregated, no personal data
Status: optional

Sparta pilot app (future)

Opt-in, pseudonymised wardrobe data
Not available until the pilot runs
Not part of this audit

Synthetic test data

Only for testing the cleaning script
Never used for results
Clearly labelled as synthetic

2 Raw zone

Raw files

Folder: data-audit/raw/
Not committed to git (size and licence)
Recorded for each file:
- source link
- date accessed
- SHA-256 checksum
- row and column counts

Trends export

data-audit/raw/trends/
Same records as above

3 Audit and cleaning

1. **Schema and type checks** — keep IDs, codes as text (leading zeros)
2. **Missing values** — blanks in FN, Active, age: fill or flag
3. **Duplicates** — repeated rows and article IDs; keep or merge
4. **Outliers** — price and age ranges; cap, flag or keep
5. **Menswear filter** — index_group_name = Menswear (confirm)
6. **Minimise** — drop hashed postal code, free text
7. **Bias and representativeness checks** — age, channel, product mix, season, brand

Each treatment is logged with its reason (clean.py)

4 Feature engineering

Feature engineering log

Each feature: logic and leakage risk

- week, month, year from date
- season from date (hemisphere)
- colour family (colour_group_name)
- pattern (graphical_appearance)
- category mapped to Sparta taxonomy (FDD section 13)
- price band
- customer age group

Leakage control: features use only information available at the time; a time-based split is used later.

5 Outputs

Cleaned dataset

version tag data-v1.0
CSV or Parquet

Data dictionary and checksum

every column; SHA-256

Audit report (PDF)

maximum 8 pages

Datasheet (appendix)

3 to 4 pages

Used by

D3: EDA notebook
D4: modelling checkpoint
Both read the cleaned dataset

Controls that apply along the whole flow

- Licence: raw H&M files stay out of git; rules say non-commercial research and education only. Check redistribution before publishing any derivative.
- Privacy: customer IDs are already hashed; no attempt to re-identify; postal code and free text are dropped; no personal data in the repository (DPDP Act, 2023 statement in the report).
- Provenance: source link, date accessed and SHA-256 checksum recorded for every raw file; the cleaning script reproduces the cleaned dataset from raw data.
- Scope: H&M is one brand's mostly European customers, so it is a stand-in for purchase demand, not for Sparta users. Representativeness checks say so.
- customers.csv has no gender field, so menswear is defined by article (index group), not by customer; some buyers of menswear are not men.
- Dashed boxes are not used in this audit: they are optional or not yet available.